

Lecture 1

Introduction to Machine Learning & Pattern Recognition

Welcome to this Course

Keji He（何科技）
Assistant Professor@SDU-AI



What emoji best describes your mood today?

Roadmap

- Introductions
- Course Goals and Plans
- What Is Machine Learning?
 - Definition
 - History
- The ML Process (Lifecycle)
- Logistics

Introductions

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Keji He (何科技)

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Assistant Professor

School of Artificial Intelligence
Shandong University

Email: keji.he@sdu.edu.cn



Biography

Dr. Keji He is an assistant professor at the School of Artificial Intelligence, Shandong University (SDU). He finished his PhD at the Institute of Automation, Chinese Academy of Sciences, supervised by Prof. [Liang Wang](#) (IEEE Fellow) and Prof. [Yan Huang](#). He is also a jointly PhD at the National University of Singapore (NUS), supervised by Prof. [Xinchao Wang](#).

His research interests include vision-and-language understanding, generative AI, and embodied AI.

Xin Xin (辛鑫)



Xin Xin (辛鑫)

Associate Professor

Shandong University, China

Email: xinxin@sdu.edu.cn

Xin Xin is now an Associate Professor in the School of Computer Science and Technology of Shandong University, as a member of the [Information Retrieval Lab](#). He also serves as the Dean Assistant for the School of Artificial Intelligence, Shandong University. Before that, he got his PhD degree in computing science from University of Glasgow, under the supervision of Prof. [Joemon Jose](#). Formerly, he got his master degree from School of Software Engineering in SJTU (上海交通大学) and his bachelor degree from XJTU (西安交通大学). His research interests span recommender systems, reinforcement learning, graph learning, causal inference for recommendation and NLP. His work appeared in several top-tier ML & IR conferences including SIGIR, WSDM, CIKM, IJCAI, ACL and UAI. His PhD is supported by Chinese Scholarship Council.

Wei Liu (刘伟)

Teaching Assistant
Graduate Student (First Year)
Major: Artificial Intelligence

Email: 17660104818@163.com
You can also reach him in the
QQ group.



MACHINE LEARNING

A
Really
Cool
and
Fun
Thing!!!



Course Goals and Plans

- Introductions
- Course Goals and Plans
- What Is Machine Learning?
 - Definition
 - History
- The ML Process (Lifecycle)
- Logistics



Why are you here? What do you want to get from this class?

By the End of Course, You Should:

- Have a **rigorous foundation** in core ML concepts and algorithms, connecting the math to the methods we use.
- Be able to **implement, train, and debug** standard models in Python/PyTorch and move from data prep to experiments.
- Be able to **design sound evaluations** and document your work so results are **reproducible and responsible**.
- Be **prepared for advanced courses and research**.

These are ambitious goals!

Accel in Advanced ML Courses

Advanced ML course:

- [Computer vision](#), [NLP](#), [RL](#), [Statistical Learning Theory](#), and Many Special Topics Courses on AI

Require understanding of basic concepts:

- Model design
- Stochastic Gradient Descent
- Maximum Likelihood
- Loss Structure
- Supervised Learning
- Unsupervised Learning
- Classification
- Regression
- Regularization and the Bias/Variance Tradeoff
- Dimensionality Reduction
- Clustering and Density Estimation
- Experiment Design and Tracking
- ...

What Is Machine Learning?

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What Is Machine Learning (ML)?

Software systems that improve (learn) through data.

Classic Example: What is Spam?

- Difficult to define
- Easy to demonstrate

Classic Example: Face detection?

- Difficult to program
- Easy to demonstrate

Basic Recipe:

- Use examples (**data**) to teach (**fit**) a **model**.
- Use the **model** to **make decisions**.

What Is Artificial Intelligence?

Artificial intelligence (AI) refers to the capability of computational systems to perform tasks typically associated with human intelligence, such as learning, reasoning, problem-solving, perception, and decision-making.

-- Wikipedia

The primary technology behind modern AI is
Machine Learning (ML)

Artificial Intelligence Is the **Goal** Machine Learning Is the **Method**

Terms are often used interchangeably (even by experts).

An entrepreneur's note on AI marketing

If you are selling

- **Machine Learning** then you call it **Deep Learning**. *(2015-2020)*
- **Deep Learning** then you call it **AI**. *(2020-2024)*
- **AI** you call it an **Agent**. *(2024-Now)*

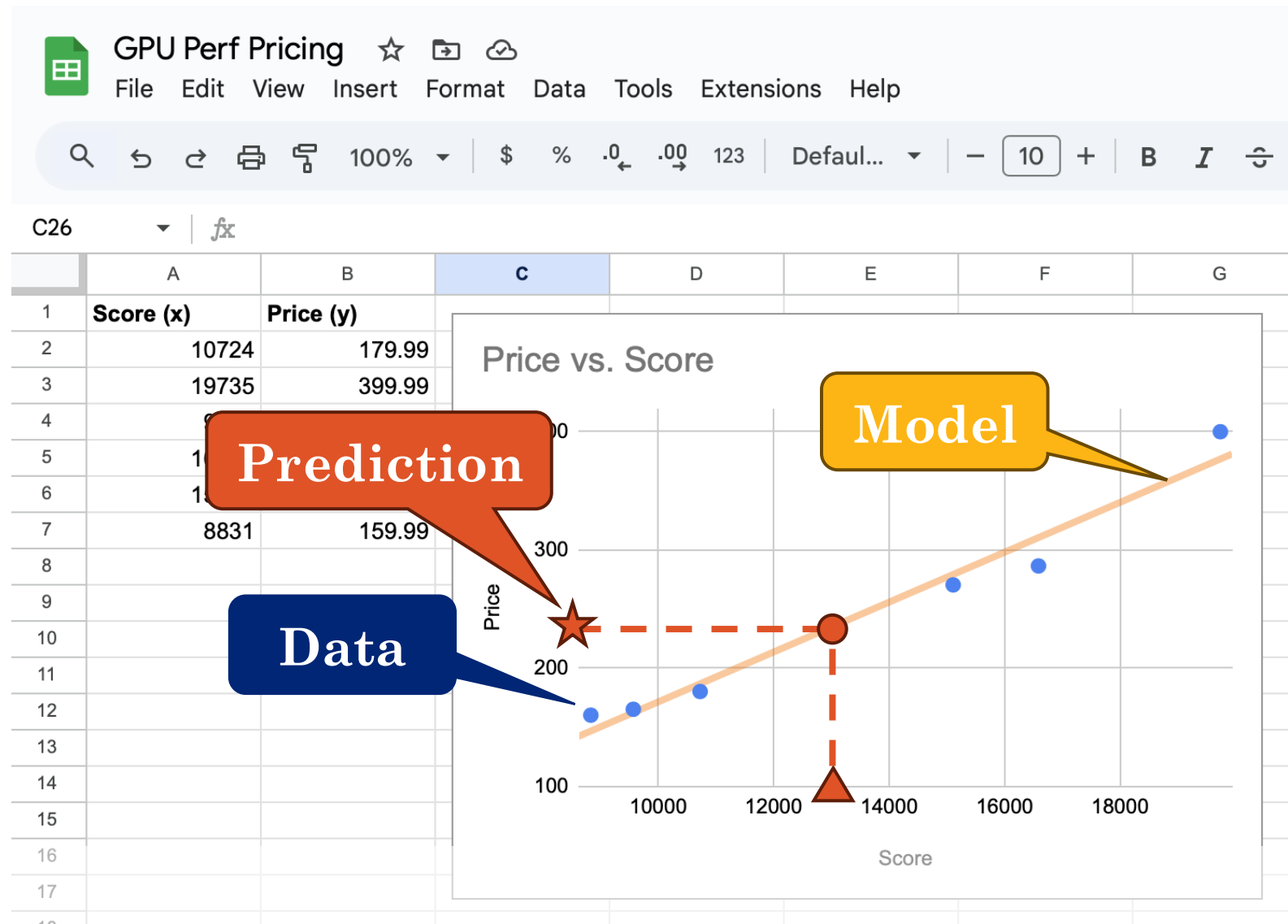
You have (probably) already
done **Machine Learning**

Linear Regression is Machine Learning

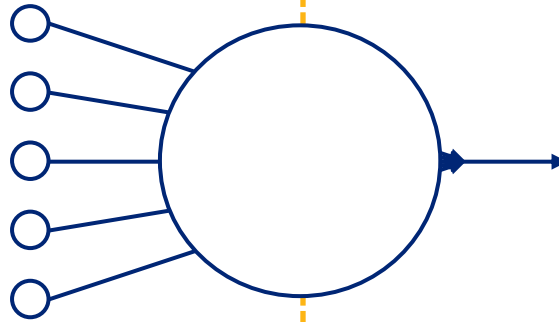
Basic Recipe:

Training: Use **data** to teach (fit) a **model**.

Inference: Use the **model** to make **predictions** (decisions).



Training



Model

Training

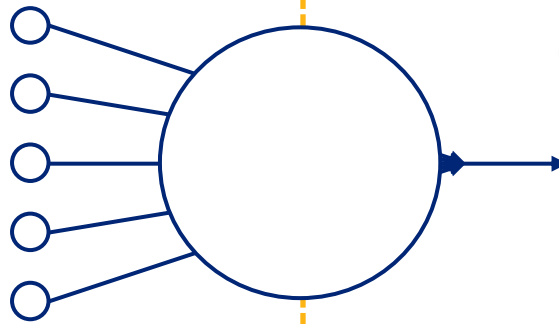
Inference



Data



Training



Model



Query



Decision



Application

When should I use Machine Learning?

What kinds of problems are
Machine Learning Problems?

Kinds of Problems

Engineering Problem: Can be solved with a direct, specifiable algorithm or a set of hand-written rules.

Machine Learning Problem: For which it is easy to demonstrate or evaluate the solution but difficult to directly implement.

A Human Problem: The problem cannot be well specified and/or human judgement is required.

→ Often require *Engineering + ML + Humans* 🙌

A Machine Learning Problem

A problem for which it is **easy to demonstrate** or **evaluate the solution** but **difficult to directly implement**.

Machine Learning Solution:

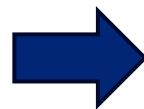
The system **learns the desired behavior** (e.g., prediction, representation, or a policy) through **demonstration** or **experience**.

Example:

Determine if a text message is spam

How do you
define Spam?

Spam is **difficult to define** and
depends on the receiver.



Easier to **demonstrate examples**
and **learn a function** to detect

Is Chatting a Machine Learning Problem?

Example (ChatGPT):

Engage a human in a productive conversation

How do you
program this?

Welcome to

```
EEEEEE LL      IIII  ZZZZZZ  AAAAA
EE      LL      II     ZZ     AA  AA
EEEEEE LL      II     ZZZ     AAAAAA
EE      LL      II     ZZ     AA  AA
EEEEEE LLLLLL  IIII  ZZZZZZ  AA  AA
```

Eliza is a mock Rogerian psychotherapist.
The original program was described by Joseph Weizenbaum in 1966.
This implementation by Norbert Landsteiner 2005.

```
ELIZA: Is something troubling you ?
YOU:   Men are all alike.
ELIZA: What is the connection, do you suppose ?
YOU:   They're always bugging us about something or other.
ELIZA: Can you think of a specific example ?
YOU:   Well, my boyfriend made me come here.
ELIZA: Is it important to you that your boyfriend made you come here ?
YOU:   He says I'm depressed much of the time.
ELIZA: I am sorry to hear that you are depressed.
YOU:   It's true. I am unhappy.
ELIZA: Can you explain what made you unhappy ?
YOU:
```

Entertaining but
**it can't truly
understand and
response.**

We can **demonstrate**
good conversations.

We can **judge** good
conversations.

ELIZA (1966)

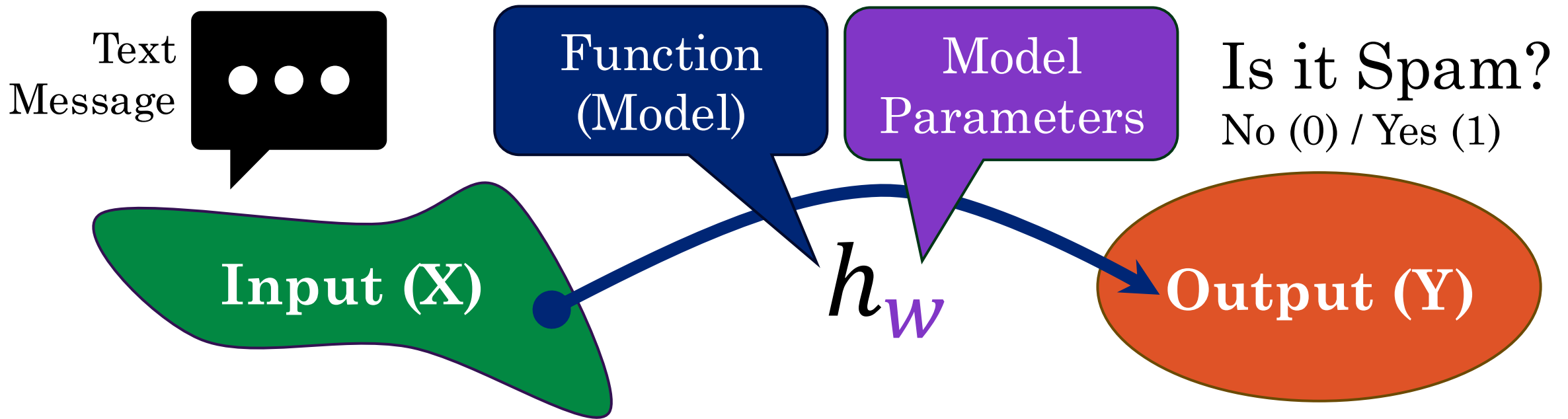
Rule based

Conversational System

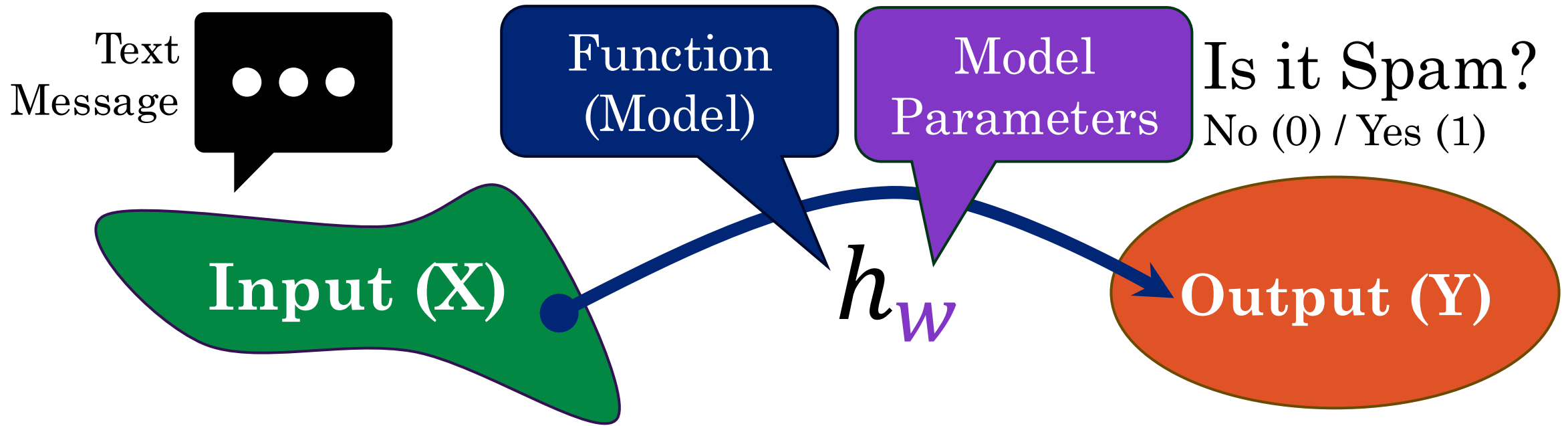
<https://en.wikipedia.org/wiki/ELIZA>

<https://web.njit.edu/~ronkowit/eliza.html>

Machine Learning as Learned Function Approximation



Machine learning becomes the process of “**learning**” the **model parameters** from data or interaction with the world.



Learning Settings

Supervised (Demonstration)	Unsupervised (Inner Pattern)	Reinforcement (Reward)
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Observe:	$\{(\mathbf{X}, \mathbf{Y})\}$	$\{\mathbf{X}\}$	$\mathbf{X}, \text{reward}(\cdot)$
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Supervised Learning

Trying to learn a relationship between observed $\{(\mathbf{X}, \mathbf{Y})\}$ pairs.

Classification



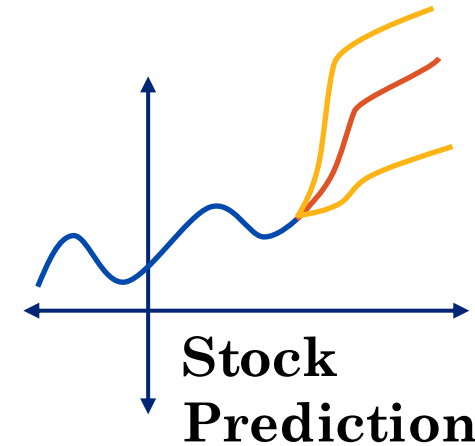
Image Labeling

$\mathbf{X}$: Image
 $\mathbf{Y}$: {Hot Dog, ...}

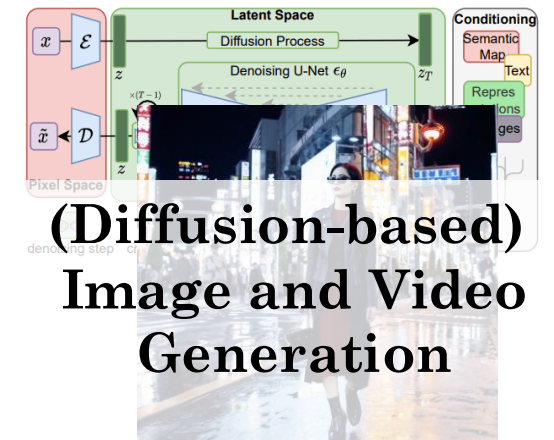


$\mathbf{X}$: Prompt
 $\mathbf{Y}$: Next Word

Regression



$\mathbf{X}$: History
 $\mathbf{Y}$: Next Value



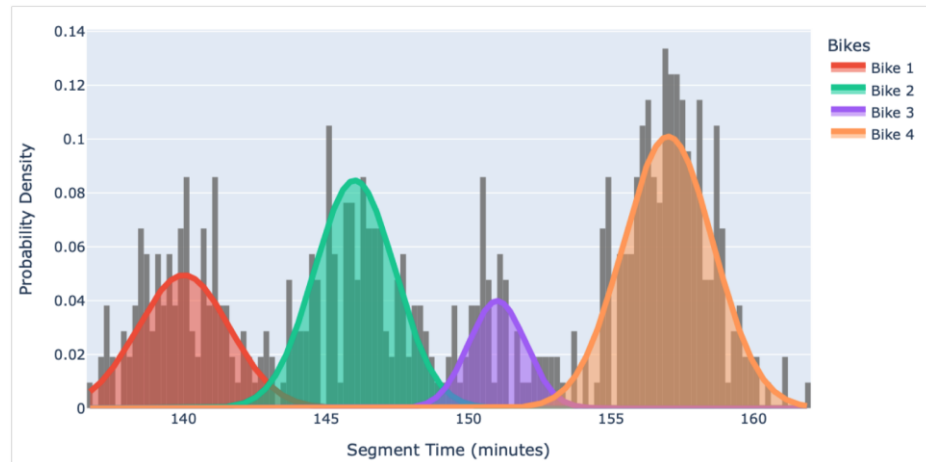
(Diffusion-based)
Image and Video
Generation

$\mathbf{X}$: Prompt+Noise
 $\mathbf{Y}$: Pixel Values

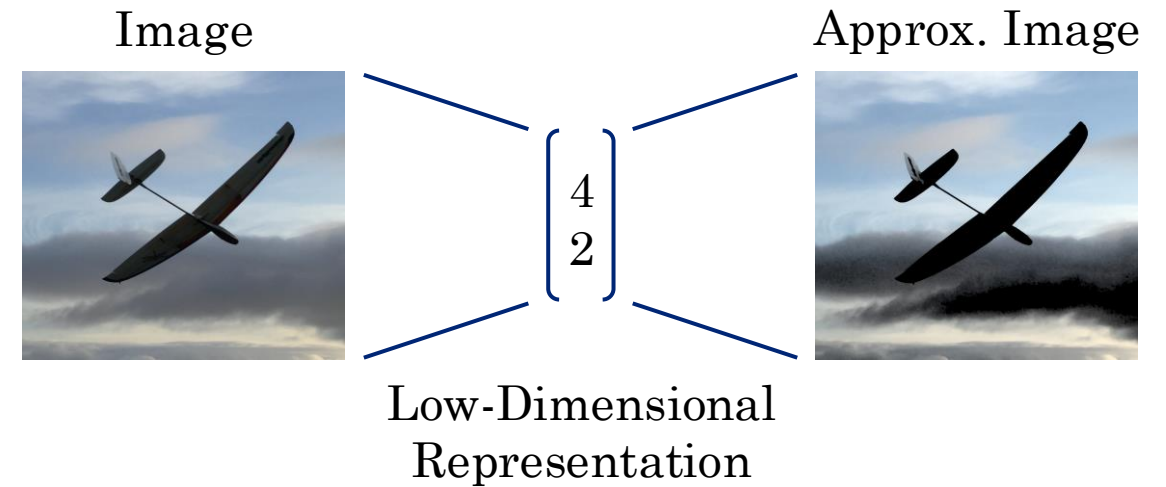
Unsupervised Learning

Trying to **model** the data in the **absence** of **explicit labels**.

Clustering & Density Estimation



Dimensionality Reduction



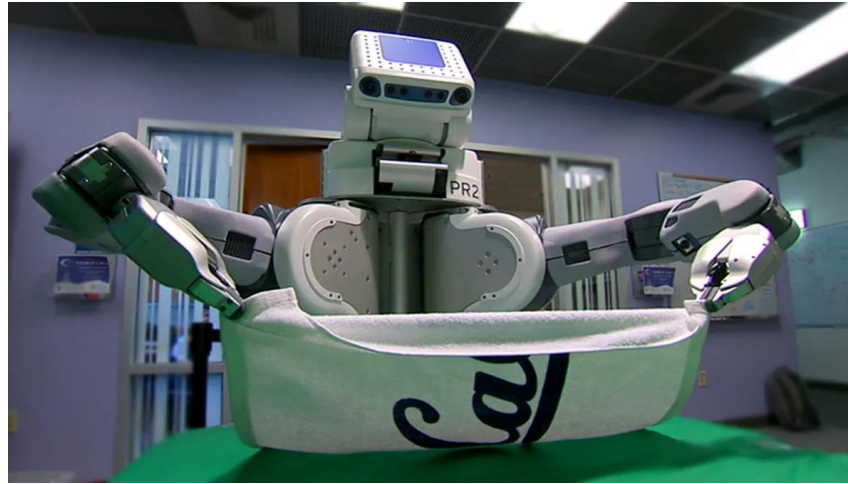
Used for **visualization** and as a step in **other ML tasks**.

Reinforcement Learning

Learning from **reward signals** often with complex **multi-step** (discrete or continuous) **action** sequences.



Action: next move,
Reward: Win/Lose



Action: change in joint angles
Reward: Fold quality



Action: next token
Reward: answer quality

History of Machine Learning

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History of ML

1950s–60s Early Days	1970s–80s Challenges and Advances	1990s Rise of Statistical ML	2000s Big Data Era	2010s Deep Learning Revolution	Present
Self-learning checkers program (1959) Perceptron (1957)	Decision trees, RL basics, and the rediscovery of NNs	Probabilistic models & statistical learning. Focus on math foundations.	Datasets grew & computation became cheaper Rise of Data Mining & Data Science	Deep learning (2012)	?
					



What do you think defines AI today?

History of ML

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Today – GenAI

What does it mean?

Models that can
generate the data.

Why is it important?

Unlocking new advanced
general AI abilities.

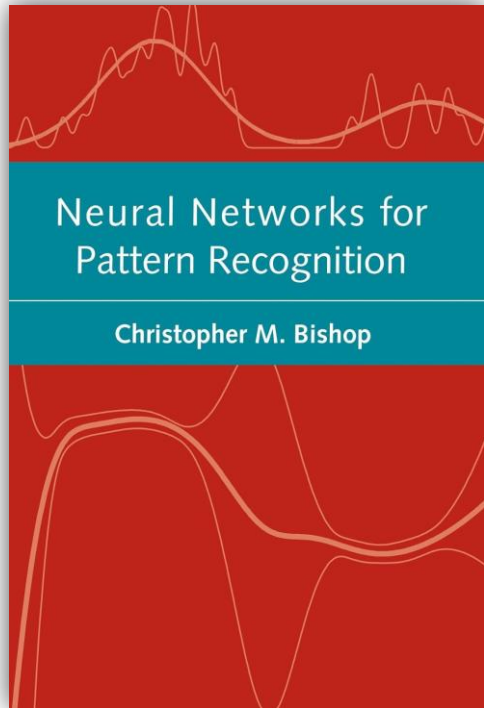
Will we cover it?

Yes!

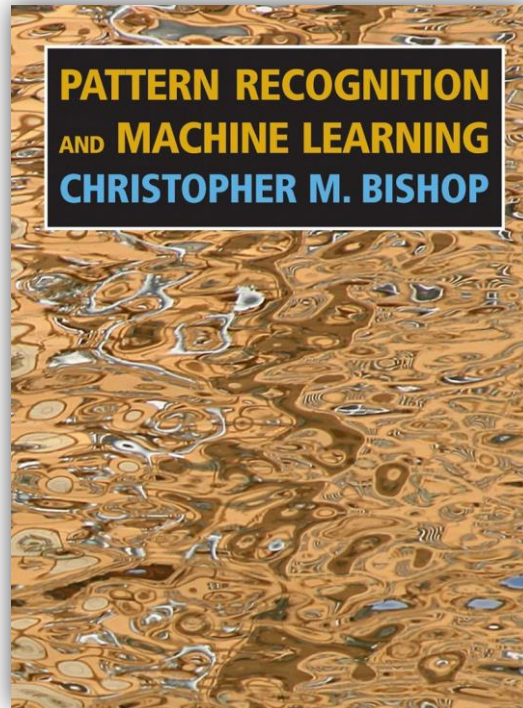
How We teach ML Has *Evolved*

Probability and Linear Algebra

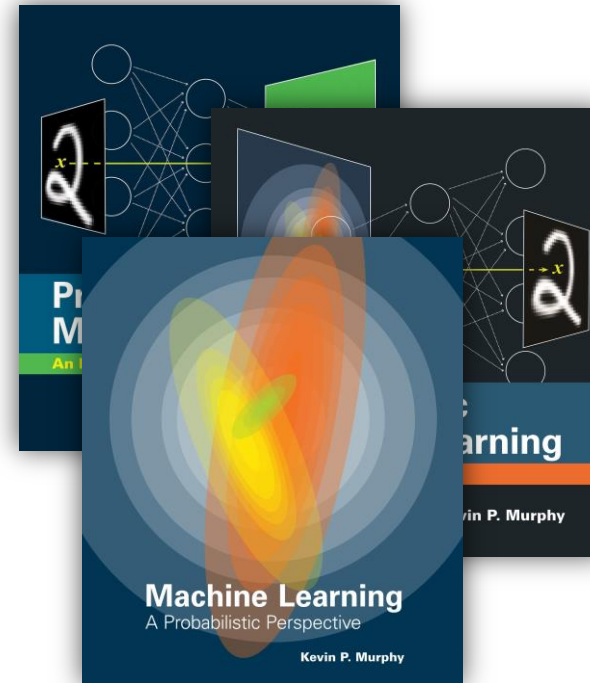
Deep Learning
(NN + Prob + Lin Alg.)



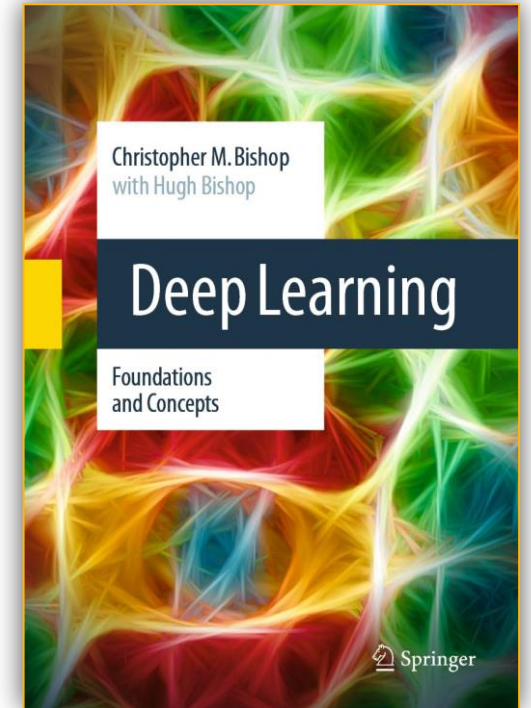
1996



2006



2013-2022

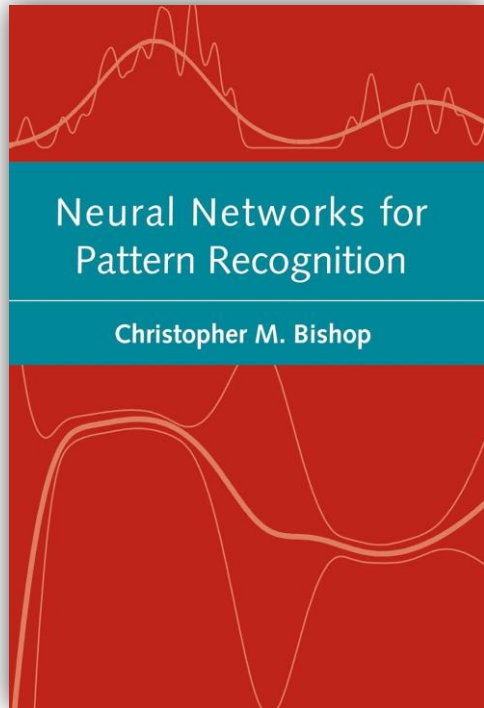


2023

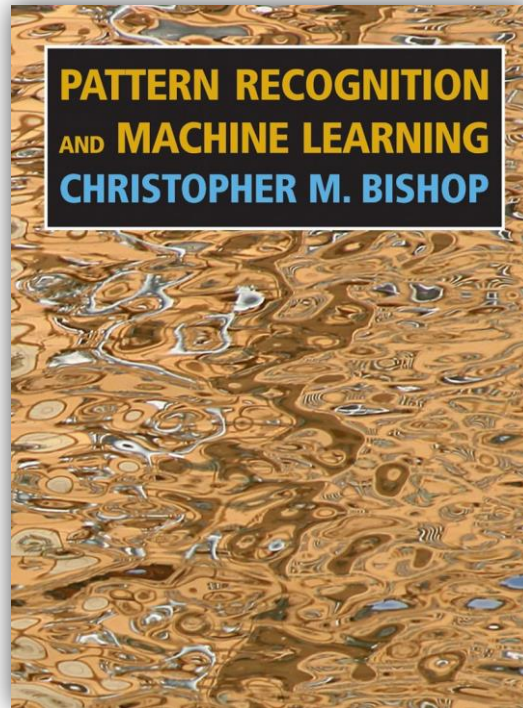
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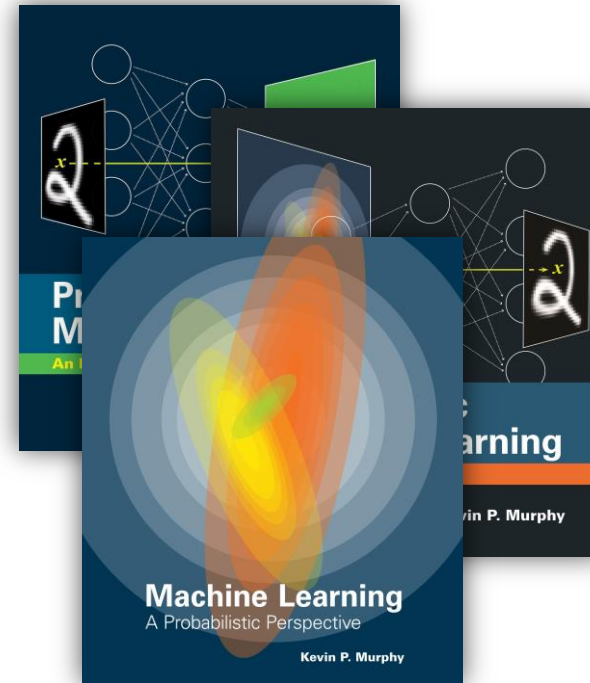
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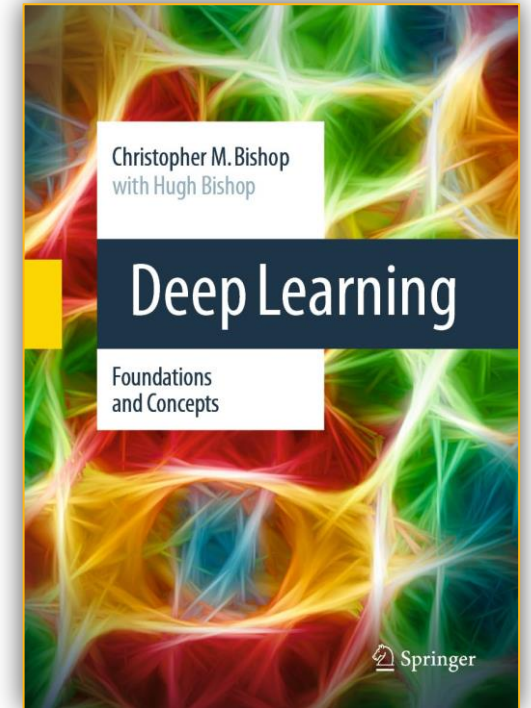
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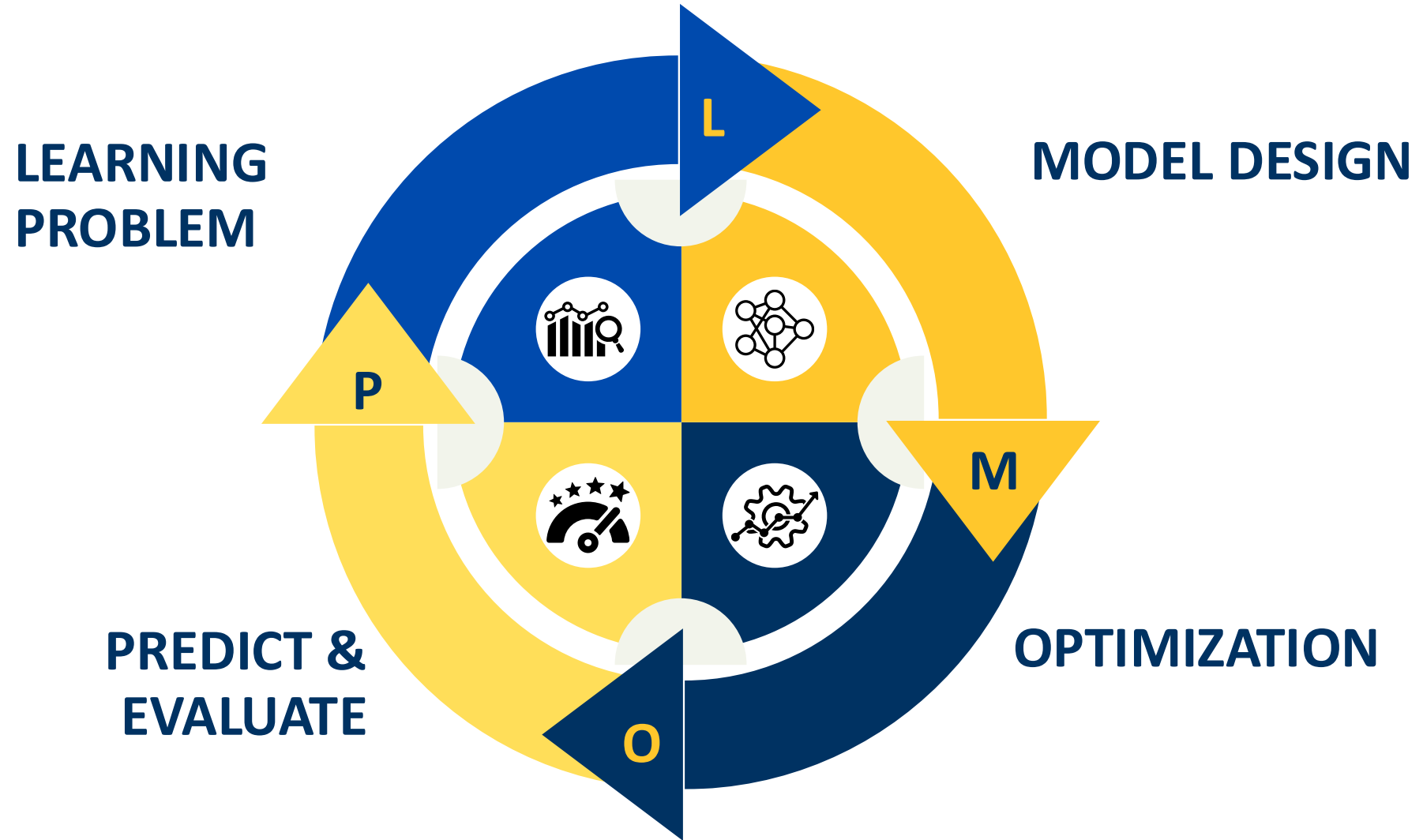


2023

The ML Process (Lifecycle)

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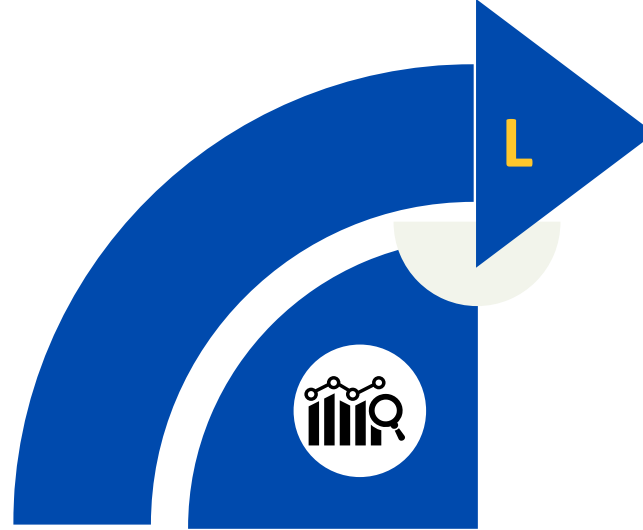
ML Lifecycle



ML Lifecycle

This stage is about framing the real-world question into something a machine learning model can answer.

LEARNING PROBLEM



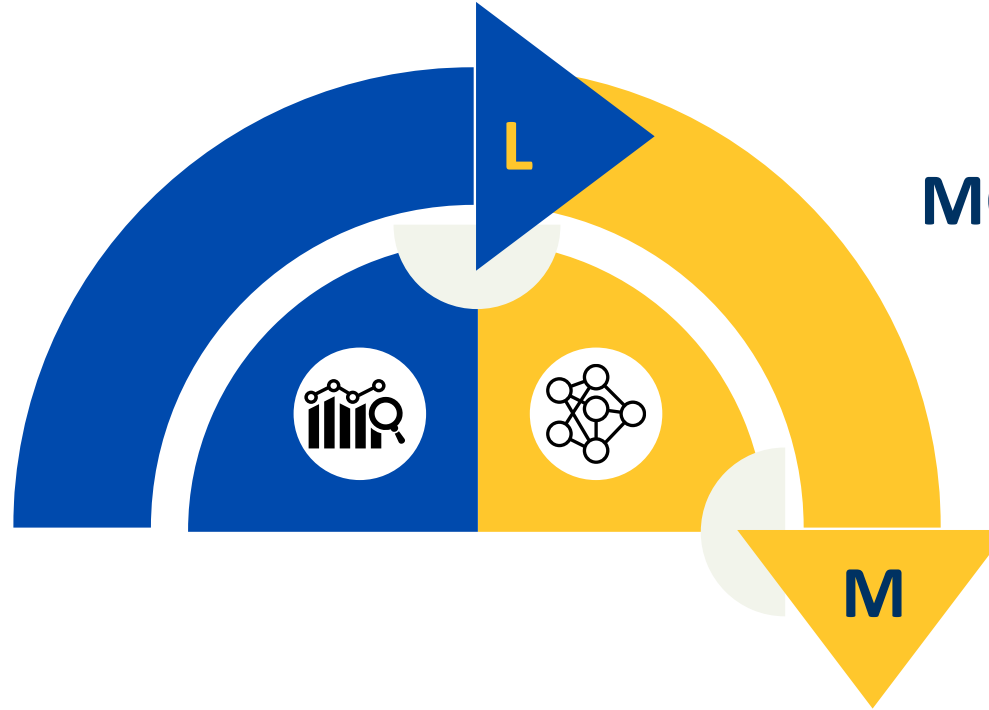
- **Target:**
 - What do I want to **predict**?
What is the machine learning task?
- **Objective:**
 - How would I **evaluate success**? What loss should I use?
- **Data:**
 - What **data** do I have?
 - Data representation ?
 - Training/Test split

ML Lifecycle

Choose and design an appropriate model.

LEARNING PROBLEM

- Target
- Objective
- Data



MODEL DESIGN

- Feature Engineering
- Model family/Architecture
- Hypothesis space
- Inductive biases / Assumptions

ML Lifecycle

Adjusting the model's parameters to minimize error using optimization algorithms.

LEARNING PROBLEM

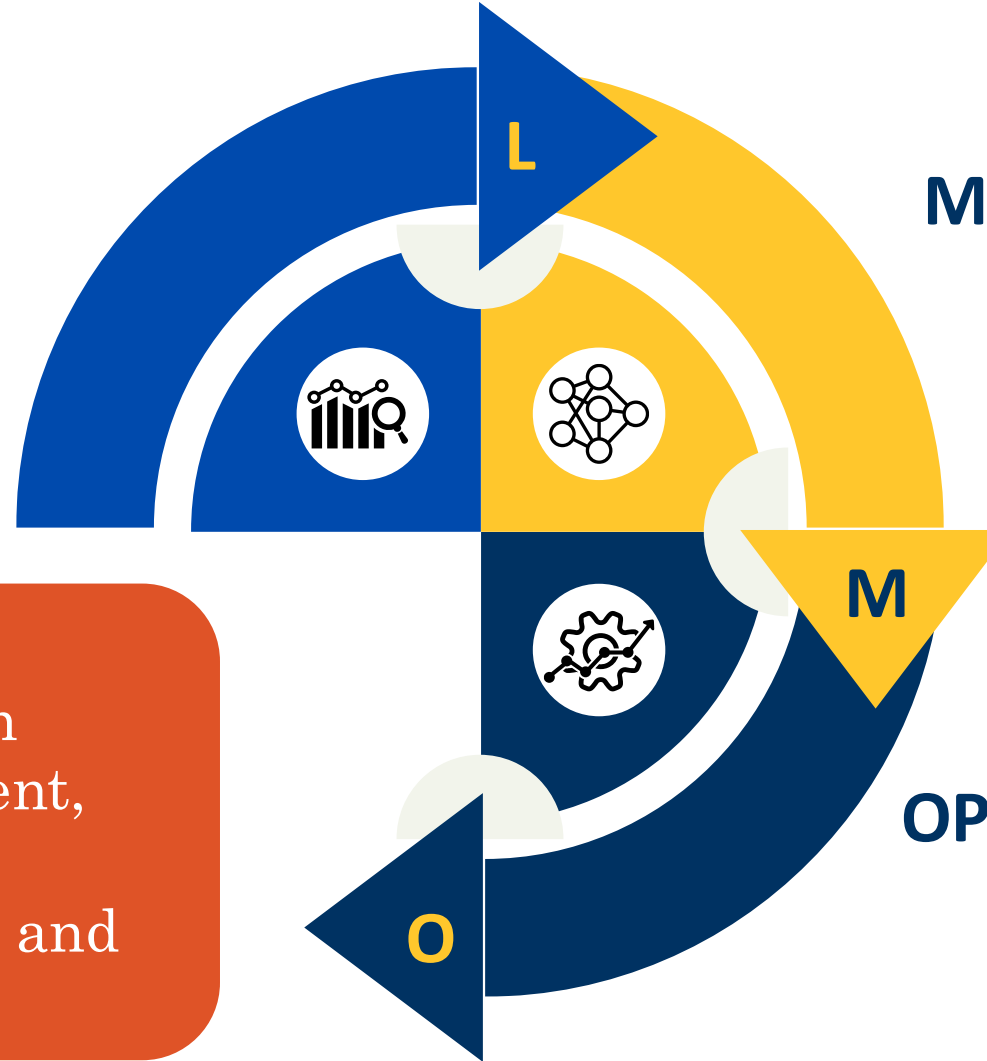
- Target
- Objective
- Data

- Define a loss
- Choose an optimization method (gradient descent, etc.)
- Manage regularization and overfitting

MODEL DESIGN

- Feature Engineering
- Model family/Architecture
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- Inductive biases / Assumptions

OPTIMIZATION



ML Lifecycle

Test how well the model performs.

LEARNING PROBLEM

- Target
- Objective
- Data

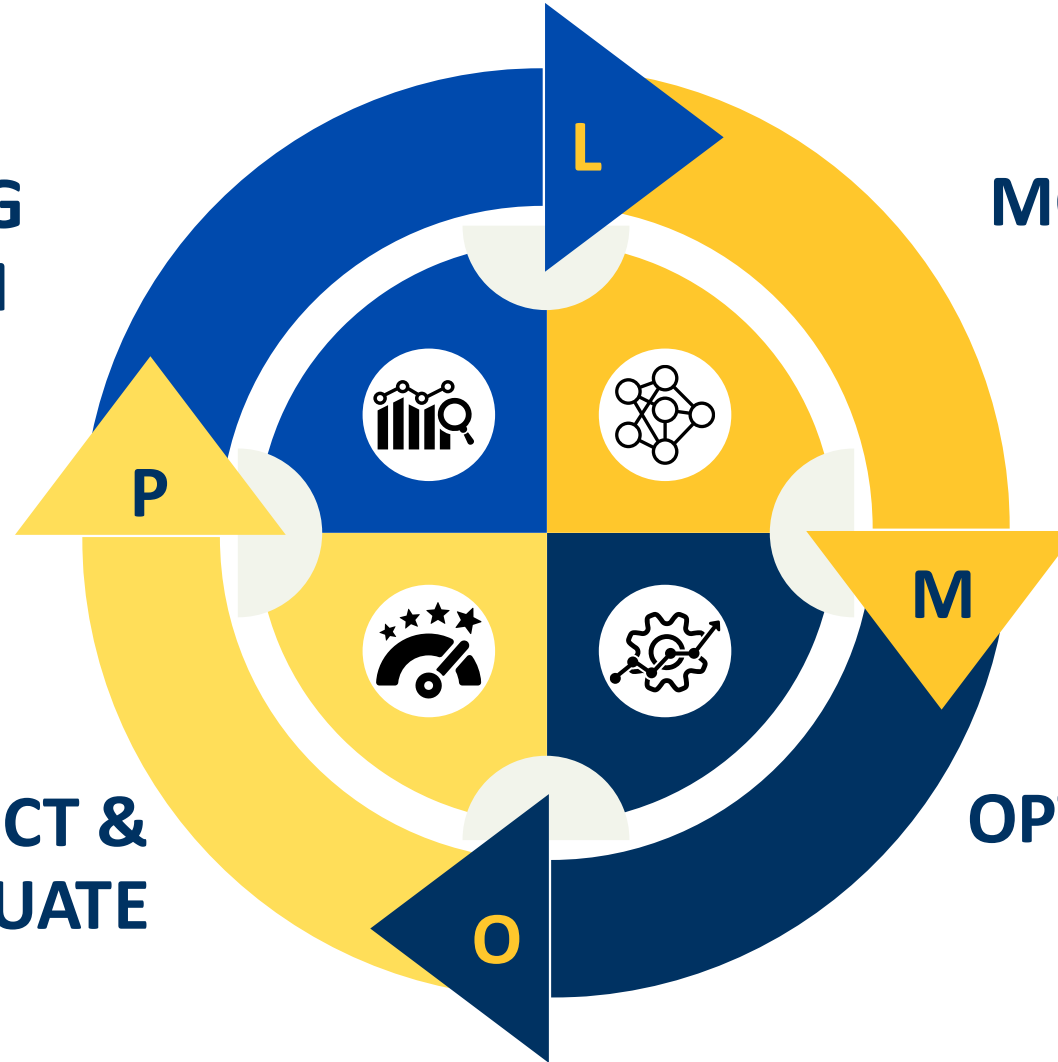
MODEL DESIGN

- Feature Engineering
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PREDICT & EVALUATE

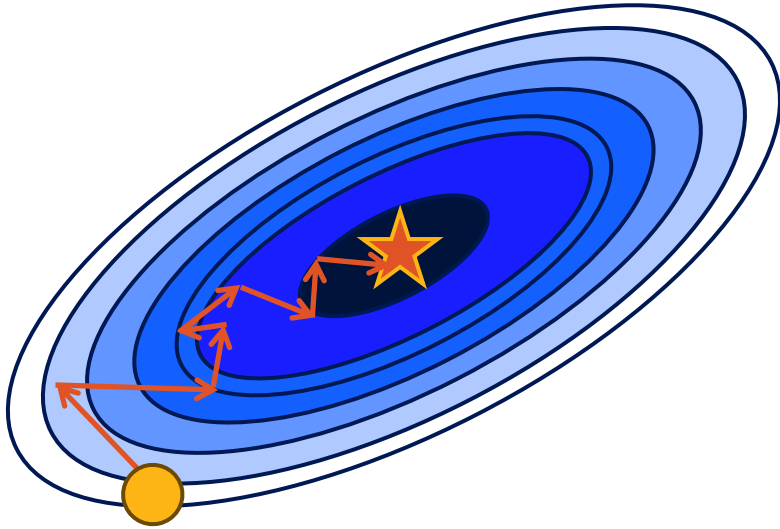
Evaluate predictions based on evaluation metrics

OPTIMIZATION

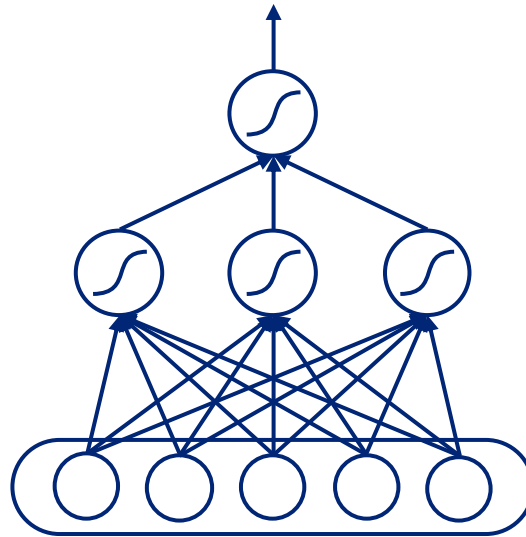


Study Pattern

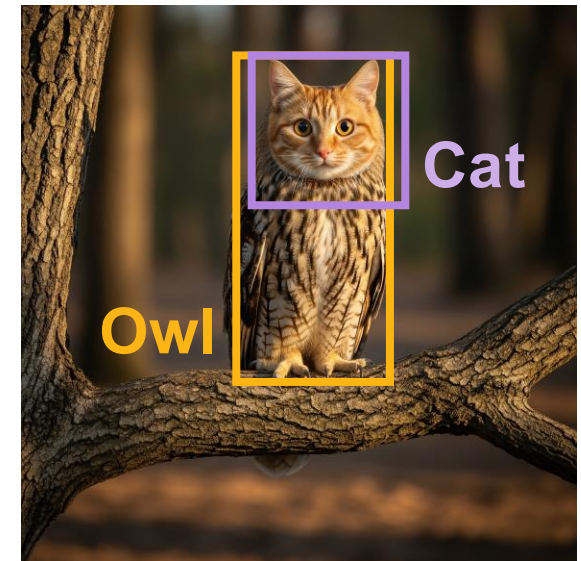
Algorithm



Model

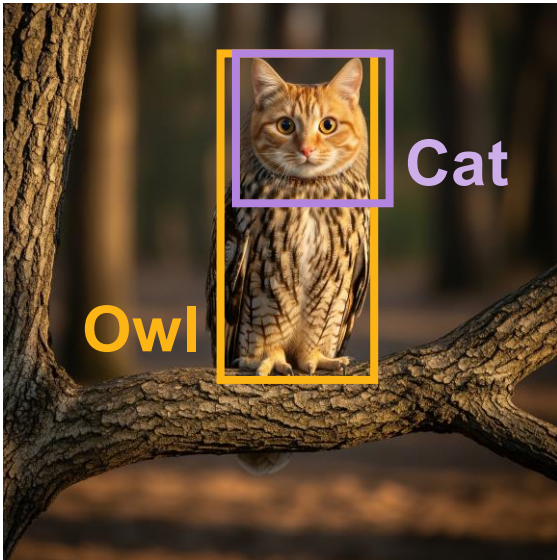


Application



Greater Focus on Application Framing

Application



- When should I use machine learning?
- How do I frame a machine learning problem?
- How do I design and train a model?
- How do I evaluate the model?

Greater Focus on ML Engineering

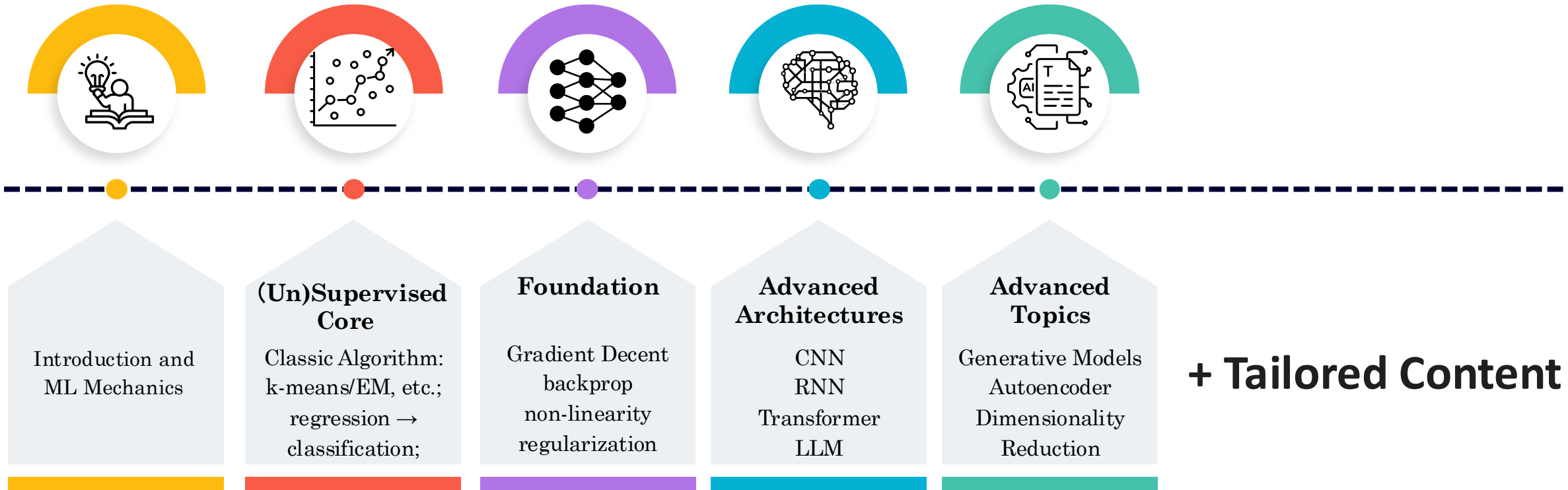
Tools for ML:

- **pandas** (<https://pandas.pydata.org/>) for data manipulation  pandas
- **Plotly** (<https://plotly.com/>) and **Matplotlib**  plotly  matplotlib (<https://matplotlib.org/>) for data visualization.
- **Scikit-learn** (<https://scikit-learn.org/stable/>) for classic machine learning tasks. 
- **PyTorch** (<https://pytorch.org/>) and **HuggingFace**  Hugging Face 
- **Weights-and-Biases** (<https://wandb.ai/site/>) for experiment management. 

Logistics

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Course Map at a Glance



Assessment

- 总分100

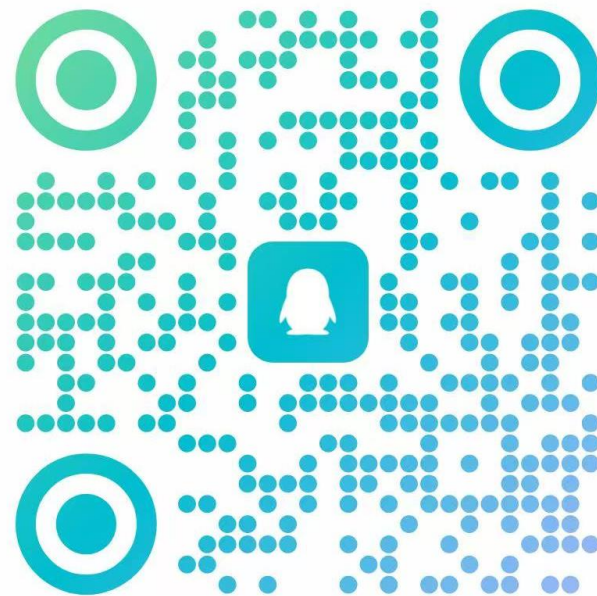
出勤 10% + 实验&报告 30% + 期末考试 60%

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加群请将昵称改为 姓名-学号 格式



机器学习与模式识别

群号: 1090319040



Prerequisites

Prerequisites

- What you need:
 - Programming skills (Python is best)
 - Implement functions and classes
 - Debug programs
 - Probability and statistics
 - Work with conditional probabilities and Bayes rule and compute expectations
 - Familiarity with uniform, Bernoulli, and Gaussian distributions
 - Linear algebra and Calculus
 - Understand eigenvectors and eigenvalues
 - Compute gradients, apply the chain rule

Lecture 1

Introduction to Machine Learning